

Linux Practical Examination

Submission Instructions

1. Perform all questions on Linux.
2. Take a screenshot of execution/output for every question.
3. Add the screenshot below the respective question.
4. Create a README.md with:
 - o Linux version
 - o Commands used
 - o Brief result/conclusion
5. Create a GitHub repository and upload:
 - o README.md
 - o All question screenshots
6. Submit the GitHub repository link to the examiner.
7. Screenshots must be from your own execution.

Section A — File & Directory Management

Q1. Basic File Operations □ Create a

directory named LinuxExam.

- Navigate into the directory.
- Create: o student.txt o course.txt o result.txt
- Display the current working directory.
- Display the contents of the directory.

```
buntu@ip-172-31-15-234:~$  
buntu@ip-172-31-15-234:~$  
buntu@ip-172-31-15-234:~$ mkdir linuxexam  
buntu@ip-172-31-15-234:~$ ls  
practice_vim  aws10  aws13  aws2  aws5  aws8      cat      demo.py  demo2    dev      linuxexam  newdemo1  practice_vim  
lpha1        aws11  aws14  aws3  aws6  aws9      demo.csv demo.txt  demo_file devops  myaws1    newdemo2  
ws1          aws12  aws15  aws4  aws7  bollywood demo.jpg  demo.txt2 demo_file2 linux    newdemo   newfile  
buntu@ip-172-31-15-234:~$ cd linuxexam  
buntu@ip-172-31-15-234:~/linuxexam$ touch student.txt course.txt result.txt  
buntu@ip-172-31-15-234:~/linuxexam$ pwd  
home/ubuntu/linuxexam  
buntu@ip-172-31-15-234:~/linuxexam$ ls  
course.txt  result.txt  student.txt  
buntu@ip-172-31-15-234:~/linuxexam$
```

Q2. File Management

Inside LinuxExam:

- Copy student.txt as student_backup.txt.
- Rename course.txt to linux_course.txt.
- Delete result.txt.
- Create directories: o Documents o Backups o Scripts
- Move student_backup.txt into Backups.
- Display the directory structure.

```
ubuntu@ip-172-31-15-234: ~/linuxexam
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$ cp student.txt student_backup.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls
course.txt  result.txt  student.txt  student_backup.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ mv course.txt linux_course.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ rm result.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls
linux_course.txt  student.txt  student_backup.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ mkdir documents backups scripts
ubuntu@ip-172-31-15-234:~/linuxexam$ ls
backups  documents  linux_course.txt  scripts  student.txt  student_backup.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ mv student backup.txt backups
mv: cannot stat 'student': No such file or directory
mv: cannot stat 'backup.txt': No such file or directory
ubuntu@ip-172-31-15-234:~/linuxexam$ mv student_backup.txt backups
ubuntu@ip-172-31-15-234:~/linuxexam$ ls
backups  documents  linux_course.txt  scripts  student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls backups
student_backup.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ tree
.
├── backups
│   └── student_backup.txt
├── documents
├── linux_course.txt
├── scripts
└── student.txt

4 directories, 3 files
ubuntu@ip-172-31-15-234:~/linuxexam$
```

Q3. File Content Operations □ Add at least 5 lines of student

information to student.txt.

- Add at least 5 lines of course information to linux_course.txt. □ Display the contents using cat.
- Display the first 3 lines using head.
- Display the last 2 lines using tail.
- Count the number of lines, words, and characters in student.txt.

```
ubuntu@ip-172-31-15-234: ~/linuxexam
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$ vim student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ cat student.txt
Name:diksha Sontakke
course:MCA
College:DAyanad Science College Latur

ubuntu@ip-172-31-15-234:~/linuxexam$ cat >linux_course.txt
Course:linux
Topic:user management
editor:vim
^C
ubuntu@ip-172-31-15-234:~/linuxexam$ ls
backups  documents  linux_course.txt  scripts  student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ cat linux_course.txt
Course:linux
Topic:user management
editor:vim
ubuntu@ip-172-31-15-234:~/linuxexam$ head -n 3 student.txt
Name:diksha Sontakke
course:MCA
College:DAyanad Science College Latur
ubuntu@ip-172-31-15-234:~/linuxexam$ tail -n 2 student.txt
College:DAyanad Science College Latur

ubuntu@ip-172-31-15-234:~/linuxexam$ wc student.txt
 4  7 71 student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$
```

Section B — Users, Groups & Permissions

Q4. User Management

- Create a user named student01.
- Set a password.
- Verify that the user exists.
- Display the user's UID and GID.
- Display the user's home directory.
- Switch to the student01 account and verify the current username.

```
ubuntu@ip-172-31-15-234: ~/linuxexam
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo useradd student01
ubuntu@ip-172-31-15-234:~/linuxexam$ id student
uid=2002(student) gid=2002(student) groups=2002(student)
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo passwd student01
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-15-234:~/linuxexam$ getent passwd student01
student01:x:2003:2009::/home/student01:/bin/sh
ubuntu@ip-172-31-15-234:~/linuxexam$ id student01
uid=2003(student01) gid=2009(student01) groups=2009(student01)
ubuntu@ip-172-31-15-234:~/linuxexam$ getent passwd student01
student01:x:2003:2009::/home/student01:/bin/sh
```

```
ubuntu@ip-172-31-15-234: ~/linuxexam
$ exit
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$ su student01
Password:
$ whoami
student01
$ exit
ubuntu@ip-172-31-15-234:~/linuxexam$ █
```

Q5. Group Management

- Create a group named linuxbatch.
- Add student01 to linuxbatch.
- Verify the groups of student01.
- Create another user named student02.
- Add student02 to linuxbatch.
- Display all members of the group.

```
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo usermod -aG linuxbatch student01
ubuntu@ip-172-31-15-234:~/linuxexam$ is student01
Command 'is' not found, but can be installed with:
sudo apt install ironseed
ubuntu@ip-172-31-15-234:~/linuxexam$ id student01
uid=2003(student01) gid=2009(student01) groups=2009(student01),2010(linuxbatch)
ubuntu@ip-172-31-15-234:~/linuxexam$ groups student01
student01 : student01 linuxbatch
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo useradd student02
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo passwd student02
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-15-234:~/linuxexam$ getent student02
Unknown database: student02
Try `getent --help' or `getent --usage' for more information.
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo usermod -aG linuxbatch student02
ubuntu@ip-172-31-15-234:~/linuxexam$ id student02
uid=2004(student02) gid=2004(student02) groups=2004(student02),2010(linuxbatch)
ubuntu@ip-172-31-15-234:~/linuxexam$ getent group linuxbatch
linuxbatch:x:2010:student01,student02
ubuntu@ip-172-31-15-234:~/linuxexam$
```

Q6. File Permissions

Create project.txt and:

- Give owner Read, Write and Execute permission.
- Give Group Read and Execute permission.
- Give Others Read permission only.
- Verify the permissions.

Expected: 754

Then perform the following:

- Change the permission to 640.
- Change the owner of the file to student01.
- Change the group ownership to linuxbatch.
- Verify the final ownership and permissions.

```
ubuntu@ip-172-31-15-234: ~/linuxexam
ubuntu@ip-172-31-15-234:~/linuxexam$ touch project.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -l
total 20
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:40 backups
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 documents
-rw-r--r-- 1 ubuntu docker  46 Sep 26 03:51 linux_course.txt
-rw-r--r-- 1 ubuntu docker   0 Sep 26 04:16 project.txt
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 scripts
-rw-r--r-- 1 ubuntu docker  71 Sep 26 03:49 student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ chmod 754 project.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -l
total 20
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:40 backups
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 documents
-rw-r--r-- 1 ubuntu docker  46 Sep 26 03:51 linux_course.txt
-rwxr-xr-- 1 ubuntu docker   0 Sep 26 04:16 project.txt
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 scripts
-rw-r--r-- 1 ubuntu docker  71 Sep 26 03:49 student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ chmod 640 project.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -l
total 20
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:40 backups
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 documents
-rw-r--r-- 1 ubuntu docker  46 Sep 26 03:51 linux_course.txt
-rw-r----- 1 ubuntu docker   0 Sep 26 04:16 project.txt
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 scripts
-rw-r--r-- 1 ubuntu docker  71 Sep 26 03:49 student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo chown student01 project.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -l
total 20
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:40 backups
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 documents
-rw-r----- 1 student01 docker  46 Sep 26 03:51 linux_course.txt
-rw-r----- 1 student01 docker   0 Sep 26 04:16 project.txt
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 scripts
-rw-r--r-- 1 ubuntu docker  71 Sep 26 03:49 student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo chgrp linuxbatch project.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -l
total 20
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:40 backups
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 documents
-rw-r----- 1 ubuntu docker  46 Sep 26 03:51 linux_course.txt
```

```

total 20
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:40 backups
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 documents
-rw-r--r-- 1 ubuntu docker 46 Sep 26 03:51 linux_course.txt
-rwxr-xr-- 1 ubuntu docker 0 Sep 26 04:16 project.txt
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 scripts
-rw-r--r-- 1 ubuntu docker 71 Sep 26 03:49 student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ chmod 640 project.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -l
total 20
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:40 backups
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 documents
-rw-r--r-- 1 ubuntu docker 46 Sep 26 03:51 linux_course.txt
-rw-r----- 1 ubuntu docker 0 Sep 26 04:16 project.txt
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 scripts
-rw-r--r-- 1 ubuntu docker 71 Sep 26 03:49 student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo chown student01 project.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -l
total 20
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:40 backups
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 documents
-rw-r--r-- 1 ubuntu docker 46 Sep 26 03:51 linux_course.txt
-rw-r----- 1 student01 docker 0 Sep 26 04:16 project.txt
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 scripts
-rw-r--r-- 1 ubuntu docker 71 Sep 26 03:49 student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo chgrp linuxbatch project.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -l
total 20
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:40 backups
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 documents
-rw-r--r-- 1 ubuntu docker 46 Sep 26 03:51 linux_course.txt
-rw-r----- 1 student01 linuxbatch 0 Sep 26 04:16 project.txt
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 03:39 scripts
-rw-r--r-- 1 ubuntu docker 71 Sep 26 03:49 student.txt
ubuntu@ip-172-31-15-234:~/linuxexam$ █

```

Q7. Permission Challenge

Create:

```
LinuxExam/  
├── public/  
├── private/  
└── shared/
```

Configure permissions so that:

- public can be accessed by everyone.
- private can be accessed only by its owner.
- shared can be accessed by the owner and group members. ☐ Verify the permissions using `ls -ld`.

```
ubuntu@ip-172-31-15-234: ~/linuxexam  
ubuntu@ip-172-31-15-234:~/linuxexam$  
ubuntu@ip-172-31-15-234:~/linuxexam$ pwd  
/home/ubuntu/linuxexam  
ubuntu@ip-172-31-15-234:~/linuxexam$ mkdir -p public private shared  
ubuntu@ip-172-31-15-234:~/linuxexam$ ls  
backups  documents  linux_course.txt  private  project.txt  public  scripts  shared  student.txt  
ubuntu@ip-172-31-15-234:~/linuxexam$ chmod 755 public  
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -ld  
drwxr-xr-x 8 ubuntu docker 4096 Sep 26 04:23 .  
ubuntu@ip-172-31-15-234:~/linuxexam$ chmod 700 private  
ubuntu@ip-172-31-15-234:~/linuxexam$ chmod 770 shared  
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -ld  
drwxr-xr-x 8 ubuntu docker 4096 Sep 26 04:23 .  
ubuntu@ip-172-31-15-234:~/linuxexam$ ls -ld public private shared  
drwx----- 2 ubuntu docker 4096 Sep 26 04:23 private  
drwxr-xr-x 2 ubuntu docker 4096 Sep 26 04:23 public  
drwxrwx--- 2 ubuntu docker 4096 Sep 26 04:23 shared  
ubuntu@ip-172-31-15-234:~/linuxexam$
```

Section C — Package Management & Services

Q8. Package Management

- Update the package repository.
- Install Apache Web Server.
- Verify that Apache is installed.
- Display the installed Apache version.
- Start the Apache service.
- Check the status of Apache.
- Configure Apache to start automatically at boot.

```
ubuntu@ip-172-31-15-234:~/linuxexam$  
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo apt update && sudo apt upgrade -y  
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease  
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease  
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease  
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease  
4 packages can be upgraded. Run 'apt list --upgradable' to see them.  
The following packages were automatically installed and are no longer required:  
  linux-aws-headers-7.0.0-1009  linux-headers-7.0.0-1009-aws  linux-main-modules-zfs-7.0.0-1009-aws  linux-tools-7.0.0-1009-aws  
  linux-aws-tools-7.0.0-1009  linux-image-7.0.0-1009-aws  linux-modules-7.0.0-1009-aws  pollinate  
Use 'sudo apt autoremove' to remove them.  
  
Not upgrading yet due to phasing:  
python3-distupgrade  python3-software-properties  software-properties-common  ubuntu-release-upgrader-core  
  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 4  
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo apt install apache2  
apache2 is already the newest version (2.4.66-2ubuntu2.4).  
The following packages were automatically installed and are no longer required:  
  linux-aws-headers-7.0.0-1009  linux-headers-7.0.0-1009-aws  linux-main-modules-zfs-7.0.0-1009-aws  linux-tools-7.0.0-1009-aws  
  linux-aws-tools-7.0.0-1009  linux-image-7.0.0-1009-aws  linux-modules-7.0.0-1009-aws  pollinate  
Use 'sudo apt autoremove' to remove them.  
  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 4  
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo apt install apache2 -v  
Server version: Apache/2.4.66 (Ubuntu)  
Server built:   2026-07-06T15:33:20  
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl start apache2  
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)  
   Active: active (running) since Sat 2026-09-26 05:47:37 UTC; 7min ago  
     Invocation: 41e92f1a139442f9843cbb421ce81021  
       Docs: https://httpd.apache.org/docs/2.4/  
    Main PID: 16942 (apache2)  
      Status: "Total requests: 1; Idle/Busy workers 100/0;Requests/sec: 0.00228; Bytes served/sec: 1 B/sec"  
        Tasks: 55 (limit: 627)  
       Memory: 5.9M (peak: 7.1M)  
         CPU: 84ms  
      CGroup: /system.slice/apache2.service  
              └─16942 /usr/sbin/apache2 -k start -DFOREGROUND  
              └─16945 /usr/sbin/apache2 -k start -DFOREGROUND  
              └─16946 /usr/sbin/apache2 -k start -DFOREGROUND  
  
Sep 26 05:47:37 ip-172-31-15-234 systemd[1]: Starting apache2.service - The Apache HTTP Server...  
Sep 26 05:47:37 ip-172-31-15-234 systemd[1]: Started apache2.service - The Apache HTTP Server.  
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl enable apache2
```

Q9. Apache Web Server Configuration

- Find the Apache document root.
- Create a custom index.html.
- Put the following information on the webpage:
 - Student Name ◦
 - Roll Number ◦ Course
 - Name ◦ "Linux Practical Examination" ◻ Start/restart Apache.
- Access the webpage using the browser or curl.
- ◻ Display the HTTP response using curl.

```
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 4
ubuntu@ip-172-31-15-234:~/linuxexam$ apache2 -v
Server version: Apache/2.4.66 (Ubuntu)
Server built: 2026-07-06T15:33:20
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl start apache2
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-09-26 05:47:37 UTC; 7min ago
     Invocation: 41e92f1a139442f9843cbb421ce81021
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 16942 (apache2)
   Status: "Total requests: 1; Idle/Busy workers 100/0; Requests/sec: 0.00228; Bytes served/sec: 1 B/sec"
      Tasks: 55 (limit: 627)
     Memory: 5.9M (peak: 7.1M)
        CPU: 84ms
    CGroup: /system.slice/apache2.service
            └─16942 /usr/sbin/apache2 -k start -DFOREGROUND
              └─16945 /usr/sbin/apache2 -k start -DFOREGROUND
                └─16946 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 26 05:47:37 ip-172-31-15-234 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 26 05:47:37 ip-172-31-15-234 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl is-enabled apache2
enabled
ubuntu@ip-172-31-15-234:~/linuxexam$
```

```
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo vim /var/www/html/index.html
ubuntu@ip-172-31-15-234:~/linuxexam$ cat /var/www/html/index.html
HTML
<!DOCTYPE html>
<html>
  <head>
    <title>Linux Examination</title>
  </head>
  <body>
    <h1>Linux Practical Examination</h1>
    <p><b>Student Name:</b>Diksha Sontakke</p>
    <p><b>Roll Number:</b>6073</p>
    <p><b>Course Name:<b>MCA</b></p>
  </body>
</html>
ubuntu@ip-172-31-15-234:~/linuxexam$
```

```

ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo vim /var/www/html/index.html
ubuntu@ip-172-31-15-234:~/linuxexam$ cat /var/www/html/index.html
HTML
<!DOCTYPE html>
<html>
  <head>
    <title>Linux Examination</title>
  </head>
  <body>
    <h1>Linux Practical Examination</h1>
    <p><b>Student Name:</b>Diksha Sontakke</p>
    <p><b>Roll Number:</b>6073</p>
    <p><b>Course Name:<b>MCA</b></p>
  </body>
</html>
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl restart apache2
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-09-26 06:38:58 UTC; 41s ago
 Invocation: 73ce6e30dfb44c5a923d7b25182b7630
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 20621 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 55 (limit: 627)
   Memory: 5.5M (peak: 5.8M)
     CPU: 62ms
   CGroup: /system.slice/apache2.service
           └─20621 /usr/sbin/apache2 -k start -DFOREGROUND
             └─20624 /usr/sbin/apache2 -k start -DFOREGROUND
               └─20625 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 26 06:38:58 ip-172-31-15-234 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 26 06:38:58 ip-172-31-15-234 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-15-234:~/linuxexam$

```

```

ubuntu@ip-172-31-15-234:~/linuxexam$ ps -p 20625 -f
UID          PID    PPID    C  TIME TTY          TIME CMD
www-data    20625    20621    0   06:38 ?        00:00:00 /usr/sbin/apache2 -k start -DFOREGROUND
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl stop apache2
Failed to stop apache2.service: Unit apache2.service not loaded.
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl stop apache2
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: inactive (dead) since Sat 2026-09-26 06:52:03 UTC; 1min 15s ago
 Duration: 13min 4.382s
 Invocation: 73ce6e30dfb44c5a923d7b25182b7630
    Docs: https://httpd.apache.org/docs/2.4/
   Process: 20621 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Process: 20857 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 20621 (code=exited, status=0/SUCCESS)
   Status: "Total requests: 3; Idle/Busy workers 100/0; Requests/sec: 0.00385; Bytes served/sec: 3 B/sec"
  Mem peak: 16.8M
    CPU: 171ms

Sep 26 06:38:58 ip-172-31-15-234 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 26 06:38:58 ip-172-31-15-234 systemd[1]: Started apache2.service - The Apache HTTP Server...
Sep 26 06:52:03 ip-172-31-15-234 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Sep 26 06:52:03 ip-172-31-15-234 systemd[1]: apache2.service: Deactivated successfully.
Sep 26 06:52:03 ip-172-31-15-234 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl start apache2
ubuntu@ip-172-31-15-234:~/linuxexam$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-09-26 06:53:41 UTC; 6s ago
 Invocation: 5a1a68c1e09c425e8e521f0735d46d75
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 20892 (apache2)
   Status: "Processing requests..."
    Tasks: 55 (limit: 627)
   Memory: 5.9M (peak: 5.9M)
     CPU: 59ms
   CGroup: /system.slice/apache2.service
           └─20892 /usr/sbin/apache2 -k start -DFOREGROUND
             └─20895 /usr/sbin/apache2 -k start -DFOREGROUND
               └─20896 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 26 06:53:41 ip-172-31-15-234 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 26 06:53:41 ip-172-31-15-234 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-15-234:~/linuxexam$

```

Section D — Processes & System Administration — 15 Marks

Q10. Process Management

Perform the following:

- Display all running processes.
- Display processes in real time using top.
- Find the PID of Apache.
- Display the process using its PID.
- Stop Apache.
- Verify that Apache has stopped.
- Start Apache again.
- Verify its status.

Students should:

- Identify the PID.
- Display process details. □ Terminate the process.
- Verify that it has stopped.

```
ubuntu@ip-172-31-15-234:~/linuxexam$ ps aux
ubuntu@ip-172-31-15-234:~/linuxexam$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1   0.3  1.2 25396 12064 ?        Ss   05:34   0:16 /usr/lib/systemd/systemd --system --deseriali
root             2   0.0  0.0      0     0 ?        S    05:34   0:00 [kthreadd]
root             3   0.0  0.0      0     0 ?        S    05:34   0:00 [pool_workqueue_release]
root             4   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-rcu_gp]
root             5   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-sync_wq]
root             6   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-kvfree_rcu_reclaim]
root             7   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-slub_flushwq]
root             8   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-netns]
root            10   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/0:0H-kblockd]
root            11   0.0  0.0      0     0 ?        I    05:34   0:01 [kworker/0:1-cgroup_free]
root            13   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-mm_percpu_wq]
root            14   0.0  0.0      0     0 ?        S    05:34   0:00 [ksoftirqd/0]
root            15   0.0  0.0      0     0 ?        I    05:34   0:00 [rcu_sched]
root            16   0.0  0.0      0     0 ?        S    05:34   0:00 [rcu_exp_par_gp_kthread_worker/0]
root            17   0.0  0.0      0     0 ?        S    05:34   0:00 [rcu_exp_gp_kthread_worker]
root            18   0.0  0.0      0     0 ?        S    05:34   0:00 [migration/0]
root            19   0.0  0.0      0     0 ?        S    05:34   0:00 [kprobe-optimizer]
root            20   0.0  0.0      0     0 ?        S    05:34   0:00 [idle_inject/0]
root            21   0.0  0.0      0     0 ?        S    05:34   0:00 [cpuhp/0]
root            22   0.0  0.0      0     0 ?        S    05:34   0:00 [cpuhp/1]
root            23   0.0  0.0      0     0 ?        S    05:34   0:00 [idle_inject/1]
root            24   0.0  0.0      0     0 ?        S    05:34   0:00 [migration/1]
root            25   0.0  0.0      0     0 ?        S    05:34   0:00 [ksoftirqd/1]
root            27   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/1:0H-kblockd]
root            28   0.0  0.0      0     0 ?        S    05:34   0:00 [kdevtmpfs]
root            29   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-inet_frag_wq]
root            30   0.0  0.0      0     0 ?        I    05:34   0:00 [rcu_tasks_kthread]
root            31   0.0  0.0      0     0 ?        I    05:34   0:00 [rcu_tasks_rude_kthread]
root            32   0.0  0.0      0     0 ?        S    05:34   0:00 [kauditd]
root            33   0.0  0.0      0     0 ?        S    05:34   0:00 [khungtaskd]
root            34   0.0  0.0      0     0 ?        I    05:34   0:00 [kworker/u8:1-events_power_efficient]
root            35   0.0  0.0      0     0 ?        S    05:34   0:00 [oom_reaper]
root            37   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-writeback]
root            38   0.0  0.0      0     0 ?        S    05:34   0:01 [kcompactd0]
root            39   0.0  0.0      0     0 ?        SN   05:34   0:00 [ksmd]
root            40   0.0  0.0      0     0 ?        SN   05:34   0:00 [khugepaged]
root            41   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-kblockd]
root            42   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-blkcg_punt_bio]
root            43   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-kintegrityd]
root            44   0.0  0.0      0     0 ?        S    05:34   0:00 [irq/9-acpi]
root            46   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-tpm_dev_wq]
root            47   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-ata_sff]
root            48   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-md_bitmap]
root            49   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-md_1bitmap_io]
root            50   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-md_1bitmap_unplug]
root            51   0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/R-edac-poller]
```

```
root 654 0.0 2.0 46820 18828 ? Ss 05:34 0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run=star-top-ct
root 660 5.0 1.5 1832820 14344 ? Ssl 05:34 3:35 /snap/amazon-ssm-agent/13349/amazon-ssm-agent
root 665 0.0 0.5 18296 5196 ? Ss 05:34 0:00 /usr/lib/systemd/systemd-logind
root 716 0.0 0.0 8280 412 ttyS0 Ss+ 05:34 0:00 /usr/sbin/agetty --noret --noclear --issue-file=/etc/issue:/etc/issue.d/issue.d
root 732 0.0 0.0 8236 444 tty1 Ss+ 05:34 0:00 /usr/sbin/agetty --noret --noclear --issue-file=/etc/issue:/etc/issue.d
root 771 0.0 1.6 126220 15588 ? Ssl 05:34 0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgr
root 1046 0.0 0.0 0 0 ? I< 05:34 0:00 [kworker/R-dio/nvme0n1p1]
ubuntu 1066 0.0 0.6 21456 6352 ? Ss 05:35 0:00 /usr/lib/systemd/systemd --user --deserialize=12
ubuntu 1068 0.0 0.2 22756 2020 ? S 05:35 0:00 (sd-pam)
syslog 3370 0.0 0.2 220304 2712 ? Ssl 05:37 0:00 /usr/sbin/rsyslogd -n -iNONE
polkitd 5258 1.8 0.7 381780 7024 ? Ssl 05:37 1:16 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
root 5277 0.0 0.6 317366 5656 ? Ssl 05:37 0:00 /usr/sbin/ModemManager
root 5527 1.8 0.9 543288 8592 ? Ssl 05:37 1:14 /usr/libexec/udisks2/udisksd
root 10185 0.0 0.2 18124 2492 ? Ss 05:40 0:00 sshd-session: ubuntu [priv]
ubuntu 10252 0.0 0.3 18380 2980 ? S 05:40 0:00 sshd-session: ubuntu@pts/3
ubuntu 10255 0.0 0.3 9280 3000 pts/3 Ss+ 05:40 0:00 -bash
root 15659 0.7 0.2 7488 2464 ? Ss 05:41 0:28 /usr/sbin/cron -f -P
root 15660 0.0 0.2 2824 1936 ? Ss 05:41 0:00 /usr/sbin/acpid
root 15667 0.0 0.2 2908 1892 ? Ss 05:41 0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root 15668 0.0 0.3 80796 3348 ? Ssl 05:41 0:01 /usr/sbin/irqbalance
root 15675 0.0 1.0 279336 9488 ? Ssl 05:41 0:00 /usr/sbin/multipathd -d -s
root 15682 0.1 1.2 50408 11560 ? S<S 05:41 0:05 /usr/lib/systemd/systemd-journald
systemd+ 15693 0.0 0.7 22272 7048 ? Ss 05:41 0:00 /usr/lib/systemd/systemd-resolved
root 15916 0.1 0.5 10716 4916 ? Ss 05:41 0:03 sshd: /usr/sbin/sshd -D -o AuthorizedKeysCommand /usr/share/ecd
root 15703 0.0 0.9 37208 8440 ? Ss 05:41 0:00 /usr/lib/systemd/systemd-udev
chrony 15710 0.0 0.4 20264 3844 ? S 05:41 0:01 /usr/sbin/chronyd -n -F 1
chrony 15713 0.0 0.2 11936 2040 ? S 05:41 0:00 /usr/sbin/chronyd -n -F 1
root 15724 0.0 0.0 0 0 ? S 05:41 0:00 [psimon]
systemd+ 15725 0.0 0.8 21192 7644 ? Ss 05:41 0:00 /usr/lib/systemd/systemd-networkd
root 15734 0.0 2.4 1886152 22588 ? Ssl 05:41 0:02 /usr/lib/snapd/snapd
www-data 16617 0.0 0.1 3752 1452 ? Ss 05:43 0:00 /usr/bin/htcacheclean -d 120 -p /var/cache/apache2/mod_cache_d
root 17141 0.0 0.0 0 0 ? I 05:49 0:00 [kworker/u8:0-events_unbound]
mysql 19787 0.7 52.7 965804 491164 ? Ssl 06:15 0:12 /usr/sbin/mysqld
root 19855 0.0 1.0 18124 9788 ? Ss 06:15 0:00 sshd-session: ubuntu [priv]
ubuntu 19911 0.0 0.7 18380 6816 ? S 06:15 0:00 sshd-session: ubuntu@pts/1
ubuntu 19912 0.0 0.5 9280 5144 pts/1 Ss 06:15 0:00 -bash
root 20185 0.0 0.0 0 0 ? S 06:17 0:00 [psimon]
root 20343 0.0 0.0 0 0 ? I 06:29 0:00 [kworker/u8:2-ext4-rsv-conversion]
root 20376 0.0 0.0 0 0 ? I 06:30 0:00 [kworker/0:0-events]
root 20377 0.0 0.0 0 0 ? I 06:30 0:00 [kworker/1:2-cgroup_free]
root 20621 0.0 0.5 9168 5528 ? Ss 06:38 0:00 /usr/sbin/apache2 -k start -DFOREGROUND
www-data 20624 0.0 0.6 1214400 6264 ? Sl 06:38 0:00 /usr/sbin/apache2 -k start -DFOREGROUND
www-data 20625 0.0 0.6 1214392 6252 ? Sl 06:38 0:00 /usr/sbin/apache2 -k start -DFOREGROUND
root 20702 0.0 0.0 0 0 ? I 06:40 0:00 [kworker/1:1-events]
root 20713 0.3 4.4 365472 41584 ? Ssl 06:44 0:00 /usr/libexec/fwupd/fwupd
root 20732 0.0 0.0 0 0 ? I 06:44 0:00 [kworker/0:2]
root 20733 0.0 0.0 0 0 ? I 06:44 0:00 [kworker/u8:3]
ubuntu 20753 0.0 0.4 9960 4124 pts/1 R+ 06:45 0:00 ps aux
```

```
ubuntu@ip-172-31-15-234:~/linuxexam$ top
top - 06:46:28 up 1:12, 2 users, load average: 0.02, 0.05, 1.23
Tasks: 122 total, 2 running, 120 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.2 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.2 st
MiB Mem : 908.7 total, 63.8 free, 776.3 used, 176.1 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 132.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
19787	mysql	20	0	965804	491164	6660	S	1.0	52.8	0:12.99	mysqld
1	root	20	0	25396	12064	6624	S	0.0	1.3	0:16.80	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-kblockd
11	root	20	0	0	0	0	I	0.0	0.0	0:01.01	kworker/0:1-cgroup_free
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.17	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:00.63	rcu_sched
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.02	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kprobe-optimizer
20	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0

```

Tasks: 122 total, 2 running, 120 sleeping, 0 stopped, 0 zombie
%cpu(s): 0.3 us, 0.2 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.2 st
MiB Mem : 908.7 total, 63.8 free, 776.3 used, 176.1 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 132.3 avail Mem

  PID USER      PR  NI   VIRT   RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
19787 mysql     20   0 965804 491164 6660 S   1.0  52.8   0:12.99 mysqld
1 root      20   0 25396 12064 6624 S   0.0   1.3   0:16.80 systemd
2 root      20   0 0 0 0 S   0.0  0.0   0:00.00 kthreadd
3 root      20   0 0 0 0 S   0.0  0.0   0:00.00 pool_workqueue_release
4 root      0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-rcu_gp
5 root      0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-sync_wq
6 root      0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
7 root      0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-slub_flushwq
8 root      0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-netns
10 root     0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/0:0H-kblockd
11 root     20   0 0 0 0 I   0.0  0.0   0:01.01 kworker/0:1-cgroup_free
13 root     0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-mm_percpu_wq
14 root     20   0 0 0 0 S   0.0  0.0   0:00.17 ksoftirqd/0
15 root     20   0 0 0 0 I   0.0  0.0   0:00.63 rcu_sched
16 root     20   0 0 0 0 S   0.0  0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
17 root     20   0 0 0 0 S   0.0  0.0   0:00.00 rcu_exp_gp_kthread_worker
18 root     rt   0 0 0 0 S   0.0  0.0   0:00.02 migration/0
19 root     20   0 0 0 0 S   0.0  0.0   0:00.00 kprobe-optimizer
20 root     -51  0 0 0 0 S   0.0  0.0   0:00.00 idle_inject/0
21 root     20   0 0 0 0 S   0.0  0.0   0:00.00 cpuhp/0
22 root     20   0 0 0 0 S   0.0  0.0   0:00.00 cpuhp/1
23 root     -51  0 0 0 0 S   0.0  0.0   0:00.00 idle_inject/1
24 root     rt   0 0 0 0 S   0.0  0.0   0:00.03 migration/1
25 root     20   0 0 0 0 S   0.0  0.0   0:00.15 ksoftirqd/1
27 root     0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/1:0H-kblockd
28 root     20   0 0 0 0 S   0.0  0.0   0:00.00 kdevtmpfs
29 root     0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-inet_frag_wq
30 root     20   0 0 0 0 I   0.0  0.0   0:00.00 rcu_tasks_kthread
31 root     20   0 0 0 0 I   0.0  0.0   0:00.00 rcu_tasks_rude_kthread
32 root     20   0 0 0 0 S   0.0  0.0   0:00.00 kauditd
33 root     20   0 0 0 0 S   0.0  0.0   0:00.00 khungtaskd
34 root     20   0 0 0 0 I   0.0  0.0   0:00.56 kworker/u8:1-events_power_efficient
35 root     20   0 0 0 0 S   0.0  0.0   0:00.00 oom_reaper
37 root     0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-writeback
38 root     20   0 0 0 0 S   0.0  0.0   0:01.66 kcompactd0
39 root     25   5 0 0 0 S   0.0  0.0   0:00.00 ksmd
40 root     39  19 0 0 0 S   0.0  0.0   0:00.00 khugepaged
41 root     0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-kblockd
42 root     0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-blkcg_punt_bio
43 root     0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-kintegrityd
44 root     -51  0 0 0 0 S   0.0  0.0   0:00.00 irq/9-acpi
46 root     0 -20 0 0 0 I   0.0  0.0   0:00.00 kworker/R-tpm_dev_wq

ubuntu@ip-172-31-15-234:~/linuxexam$ pid of apache2
Command 'pid' not found, but there are 26 similar ones.
ubuntu@ip-172-31-15-234:~/linuxexam$ pidof apache2
20625 20624 20621
ubuntu@ip-172-31-15-234:~/linuxexam$ pgrep apache2
20621
20624
20625
ubuntu@ip-172-31-15-234:~/linuxexam$ pidof apache2
20625 20624 20621
ubuntu@ip-172-31-15-234:~/linuxexam$ ps -p 20625 -f
UID      PID    PPID  C  STIME TTY          TIME CMD
www-data 20625 20621 0 06:38 ?        00:00:00 /usr/sbin/apache2 -k start -DFOREGROUND
ubuntu@ip-172-31-15-234:~/linuxexam$

```

Section E — Searching, Filtering & Logs

Q11. Search and Text Processing

Inside LinuxExam:

- Create a file named `students.txt` containing at least 10 student records.
- Use `grep` to find students matching a specified name or pattern.
- Use `grep` with case-insensitive searching.
- Count matching records.
- Sort the student records.
- Search for a particular file using `find`.

Find all students whose name contains the letter a.

or:

Find all `.txt` files inside the `LinuxExam` directory.

```
ubuntu@ip-172-31-15-234:~/linuxexam$ cat >students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
107 shruti 38
108 tanvi 36
109 rutuja 34
110 nikita 30^C
ubuntu@ip-172-31-15-234:~/linuxexam$ cat students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
```

```
ubuntu@ip-172-31-15-234:~/linuxexam$ cat >students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
107 shruti 38
108 tanvi 36
109 rutuja 34
110 nikita 30
ubuntu@ip-172-31-15-234:~/linuxexam$ cat students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
107 shruti 38
108 tanvi 36
109 rutuja 34
ubuntu@ip-172-31-15-234:~/linuxexam$ grep 'a' students.txt
```

```
ubuntu@ip-172-31-15-234:~/linuxexam$ cat >students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
107 shruti 38
108 tanvi 36
109 rutuja 34
110 nikita 30
ubuntu@ip-172-31-15-234:~/linuxexam$ cat students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
107 shruti 38
108 tanvi 36
109 rutuja 34
ubuntu@ip-172-31-15-234:~/linuxexam$ grep 'a' students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
108 tanvi 36
109 rutuja 34
ubuntu@ip-172-31-15-234:~/linuxexam$ ^[[200~grep -i 'a' students.txt
grep: command not found
ubuntu@ip-172-31-15-234:~/linuxexam$ grep -i 'a' students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
108 tanvi 36
109 rutuja 34
```

```
ubuntu@ip-172-31-15-234:~/linuxexam$ cat >students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
107 shruti 38
108 tanvi 36
109 rutuja 34
110 nikita 30^C
ubuntu@ip-172-31-15-234:~/linuxexam$ cat students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
107 shruti 38
108 tanvi 36
109 rutuja 34
ubuntu@ip-172-31-15-234:~/linuxexam$ grep 'a' students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
108 tanvi 36
109 rutuja 34
ubuntu@ip-172-31-15-234:~/linuxexam$ ^[[200~grep -i 'a' students.txt
grep: command not found
ubuntu@ip-172-31-15-234:~/linuxexam$ grep -i 'a' students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
108 tanvi 36
109 rutuja 34
ubuntu@ip-172-31-15-234:~/linuxexam$ ^[[200~grep -i 'a' students.txt
grep: command not found
ubuntu@ip-172-31-15-234:~/linuxexam$ grep -i 'a' students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
108 tanvi 36
109 rutuja 34
ubuntu@ip-172-31-15-234:~/linuxexam$ grep -c 'a' students.txt
8
ubuntu@ip-172-31-15-234:~/linuxexam$ sort students.txt
101 diksha 50
102 shraddha 48
103 smaruddhi 47
104 apurvaa 43
105 shravani 42
106 anuja 40
107 shruti 38
108 tanvi 36
109 rutuja 34
ubuntu@ip-172-31-15-234:~/linuxexam$ find ~/LinuxExam -type f -name "*.txt"
find: '/home/ubuntu/LinuxExam': No such file or directory
ubuntu@ip-172-31-15-234:~/linuxexam$
```

Section F — Networking

Q12. Linux Networking

Perform the following:

- Display the system hostname.
- Display the IP address.
- Display network interfaces.
- Display the routing table.
- Test connectivity to another system or a specified website.
- Use ping to test connectivity.
- Use curl to access a webpage.
- Find the IP address of a domain name.
- Display listening ports.

```
ubuntu@ip-172-31-15-234:~/linuxexam$ hostname
ubuntu@ip-172-31-15-234:~/linuxexam$ hostname
ip-172-31-15-234
ubuntu@ip-172-31-15-234:~/linuxexam$ hostname -I
172.31.15.234
ubuntu@ip-172-31-15-234:~/linuxexam$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:3d:af:61:b3:3b brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    inet 172.31.15.234/20 metric 100 brd 172.31.15.255 scope global dynamic ens5
        valid_lft 3340sec preferred_lft 3340sec
    inet6 fe80::83d:aff:fe61:b33b/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
ubuntu@ip-172-31-15-234:~/linuxexam$ ip route
default via 172.31.0.1 dev ens5 proto dhcp src 172.31.15.234 metric 100
172.31.0.0/20 dev ens5 proto kernel scope link src 172.31.15.234 metric 100
172.31.0.1 dev ens5 proto dhcp scope link src 172.31.15.234 metric 100
172.31.0.2 dev ens5 proto dhcp scope link src 172.31.15.234 metric 100
ubuntu@ip-172-31-15-234:~/linuxexam$ ping -c 4 google.com
PING google.com (192.178.174.102) 56(84) bytes of data:
64 bytes from lcbomo-in-f102.1e100.net (192.178.174.102): icmp_seq=1 ttl=113 time=1.86 ms
64 bytes from lcbomo-in-f102.1e100.net (192.178.174.102): icmp_seq=2 ttl=113 time=1.88 ms
64 bytes from lcbomo-in-f102.1e100.net (192.178.174.102): icmp_seq=3 ttl=113 time=1.84 ms
64 bytes from lcbomo-in-f102.1e100.net (192.178.174.102): icmp_seq=4 ttl=113 time=1.83 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.827/1.850/1.877/0.018 ms
ubuntu@ip-172-31-15-234:~/linuxexam$ curl https://www.google.com
<doctype html>-html itemscope="" itentype="http://schema.org/WebPage" lang="en-IN"><head><meta content="text/html; charset=UTF-8" http-equiv="Co
ntent-Type"><meta content="/images/branding/googlelog/ix/googlelog_standard_color_128dp.png" itemprop="image"><title>Google</title><script nonce="8XR
R6ox6RHpF8u8CY5ue-Q">(function(){var _g={key:"13G3aqlucJTisemp2_P3sAg",kEXPI:"0.4318858.16105.344796.301150.5231650.595.282.79.247.5991062.9.68.3
0819226.25381059.65176.138338.37805.148747.3050.2.23178.23591.46337.32563.6313.124.9742.2646.4541.35593.2.1.1515.3355.2.4258.25165.5089.3027.19.7
76.4081.4.10543.20745.1250.2116.3221.2891.2.12.16.28227.5.964.2.147.34657.2.6335.8497.5593.239.2.18519.6293.10880.264.12284.2323.353.10681.12246.
8.7179.3933.4.21020331.3.10051.3.17972.3.5865.6506329.6.8302.8374.3.4225.3.1567.223.3.1242.1069.571.3.497.2008.1819.3.3065.156.3643.70530.1889659
.1836347.13803371.1224212.279455.1.529043.47949.3.196166.388.18774.160137.10.2.1.1.93493.164184.12.1.341898.3.12.1.2779.6777.7.7.7.9216.7629.8.12
39.1109.4895.5930.4388.5.13.5.4245.22904.5.7854.4.4988.5.4025.2.1.8.6832.7936.3396.1864.1734.7889.3094.4.2726.1859.911.4325.1500.1530.15059.2.1.8
9727.4.84.8014.15.19.5517.12034.2398.4.1484.1423.6903.8673.4.56.278.757.5.58.4.303.3011.2625.1671.4.368.4.426.4.3196.5.259.2.1.531.4331.221.5.14
5567.5.3325.886.2648.3.5269.1.1.3.14.1.4.1.3744.5.4786.1245.2.1.16.252.3527.4.131.4.480.4.2855.4.2103.2.453.5.6603.4.5563.4.933.4.69.5.7851.2879
.5.1253.5.555.98.4.3797.276.4.191.3493.5.378.1.2711.1.563.5.1144.4.145.5.5728.256.4.5848.4590.2.2695.1070.5.2750.4.1566.203.4.2410.418.1712.2.281
4.4.2047.4.1478.4.1122.2.858.1001.4.2698.4.1410.6023.5.2587.7.143.623.4626.804.395.4.571.236.501.5.578.3.2.2.2.381.3.2.2.2.6089.3.89.7.716.590.326
4.2263.4.435.1.2.3607.501.2405.438.2268.3.2.2.2.67.4.7.1039.2878.3.2.2.2.43.83.4.18.96.2300.335.212.1871.2121.4.2726.81.3966.4.512.4.1410.18.28.
```

Challenge:

Apache is installed but the webpage cannot be accessed. Identify whether Apache is running and determine whether it is listening on the expected HTTP port.